M.S. NAZ HIGH SCHOOL

Excellence in Matriculation & Academic Leadership | Est. 1989 | BISE Gujranwala Code: 112199

Single National Curriculum (SNC) & PECTAA Standards Aligned - Session 2026-2027

CLASS 10 (10TH MATRIC) - BIOLOGY

Official Syllabus, Board Paper Scheme & High-Yield Examination Revision Notes

Group: Science Group | Total Marks: 75 Marks (Theory: 60 | Practical: 15) | Time: 2 Hours 45 Minutes

SECTION 1: BISE GUJRANWALA EXAMINATION BLUEPRINT & WEIGHTAGE

- **Section A: Objective (MCQs):**
  12 Compulsory MCQs (12 Marks).
- **Section B: Short Questions:**
  Attempt 15 out of 24 Short Questions (Q2, Q3, Q4: 5/8 each = 30 Marks).
- **Section C: Long Questions:**
  Attempt 2 out of 3 Long Questions (9 Marks each = 18 Marks).

SECTION 2: CHAPTER-WISE HIGH-YIELD REVISION NOTES & CORE SLOs

Chapter 10: Gaseous Exchange:

Gaseous exchange in plants (Stomata, Lenticels), Human Respiratory System (Nasal cavity, Pharynx, Larynx, Trachea, Bronchi, Alveoli), Breathing mechanism (Inhalation vs Exhalation), Respiratory disorders: Bronchitis, Emphysema, Pneumonia, Asthma, Lung cancer, Smoking effects.

Chapter 11: Homeostasis:

Homeostasis in plants (Hydrophytes, Xerophytes, Halophytes), Human Urinary System (Kidney structure, Nephron anatomy: Bowman capsule, Glomerulus, Loop of Henle), Urine formation, Kidney disorders (Stones, Renal failure), Dialysis (Hemodialysis vs Peritoneal dialysis).

Chapter 12: Coordination and Control:

Nervous vs Chemical coordination, Components of coordinated action, Human Nervous System: Central (Brain, Spinal Cord) and Peripheral, Neurons (Sensory, Motor, Interneurons), Reflex Arc, Human Eye and Ear anatomy, Endocrine System: Pituitary, Thyroid, Adrenal, Pancreas.

Chapter 13: Support and Movement:

Human Skeleton (Axial: 80 bones vs Appendicular: 126 bones), Bone vs Cartilage, Joints (Ball and Socket, Hinge), Muscles: Skeletal, Smooth, Cardiac, Antagonism (Biceps & Triceps), Skeletal disorders (Osteoporosis, Arthritis).

Chapter 14: Reproduction:

Asexual Reproduction: Binary fission, Budding, Spore formation, Vegetative propagation, Sexual Reproduction in Plants: Flower structure, Pollination (Self vs Cross), Double Fertilization, Human Reproduction: Male & Female systems, AIDS.

Chapter 15: Inheritance:

Chromosomes and Genes, Watson-Crick model of DNA, DNA Replication, Mendel Laws: Law of Segregation, Law of Independent Assortment (Monohybrid vs Dihybrid cross), Incomplete Dominance and Co-dominance (ABO blood groups), Natural Selection.

Chapter 16: Man and His Environment:

Ecosystem components: Biotic (Producers, Consumers, Decomposers) and Abiotic, Food Chain and Food Web, Ecological Pyramids, Carbon and Nitrogen cycles, Symbiosis (Parasitism, Mutualism, Commensalism), Pollution and Conservation.

Chapter 17: Biotechnology:

Definition and scope, Fermentation (Alcoholic and Lactic acid), Fermenter types (Batch vs Continuous), Genetic Engineering: Steps in gene cloning (Restriction enzymes, Vector/Plasmid, Recombinant DNA), Applications: Human insulin, Interferon.

Chapter 18: Pharmacology:

Medicinal drugs vs Addictive drugs, Sources of drugs (Plants, Animals, Minerals, Microorganisms, Synthetic), Antibiotics (Penicillin, Cephalosporin), Antibiotic resistance, Sedatives, Narcotics, Hallucinogens.

SECTION 3: MSNS SENIOR FACULTY BOARD EXAMINATION STRATEGY

*** Draw and label Nephron anatomy and Human Brain structures with sharp pencils and precise labels.**

*** Show clear Punnett Squares for Mendel monohybrid and dihybrid genetic crosses.**